Contact guidance of single cells on adhesive lines

H. Suresh[a], A.B.C. Buskermolen[b], S. Shishvan[a,c], A. Vigliotti[a,d], A. DeSimone[e], N.A. Kurniawan[b], C.V.C. Bouten[b] and V.S. Deshpande[a]

[a] *Department of Engineering, University of Cambridge, Cambridge CB2 1PZ, U.K.*
[b] *Department of Biomedical Engineering, Eindhoven University of Technology, PO Box 513, 5600 Eindhoven, MB, The Netherlands.*
[c] *Department of Structural Engineering, University of Tabriz, Tabriz, Iran.*
[d] *Innovative Materials Laboratory, Italian Aerospace Research Centre, Capua 81043, Italy.*
[e] *SISSA – International School for advanced Studies, Trieste 34136, Italy.*

Introduction

Cell polarization, as induced by anisotropic substrate features, is an essential step in the organization, differentiation, and migration of adherent cells. However, the mechanism behind cell polarization remains a mystery. Here, we propose a new mechanism for contact guidance by investigating cells confined to micro-patterned lines of fibronectin.

Methods

Experiments

We analyse the response of Human Vena Saphena Cells seeded on a substrate micro-printed with fibronectin lines of different widths (Fig. 1a). Experiments are conducted such that cells remain isolated (no cell-cell contact) and confined within the adhesive line at all times. A homogeneous pattern of fibronectin printed on a flat PDMS surface was used as the control substrate. Focal adhesions, actin cytoskeleton and the nucleus were stained for visualization. The images of the cells acquired from an inverted microscope were fitted with an ellipse to quantify observables such as orientation and aspect ratio.

Modelling

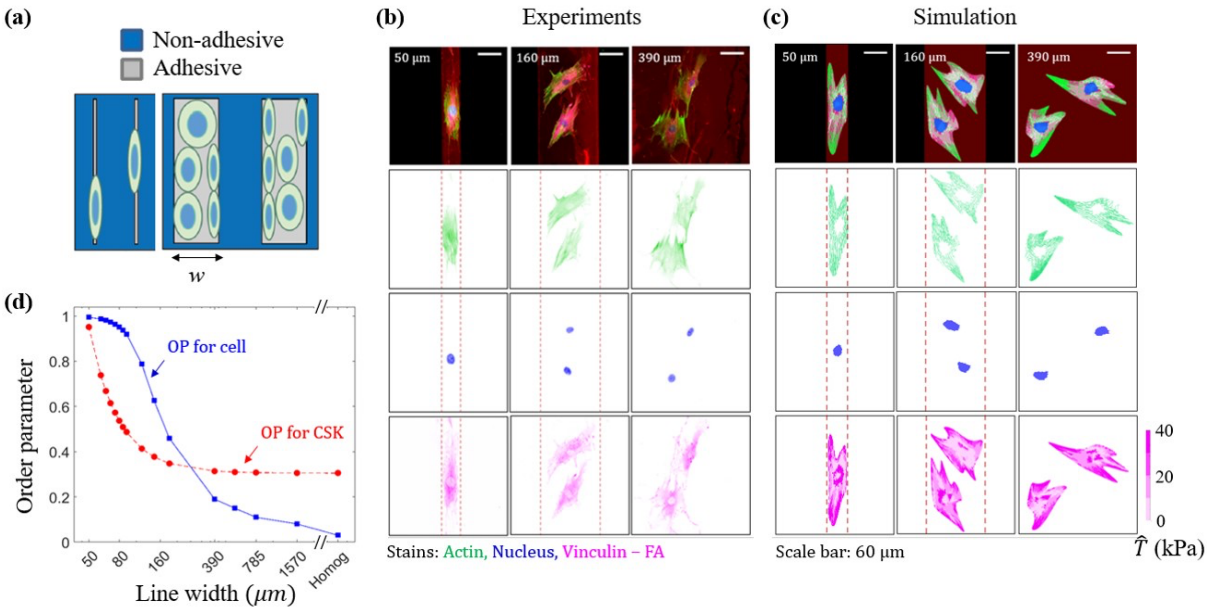
Consider a thermodynamic system comprising of an adherent cell on a substrate, immersed in a thermo-nutrient reservoir. We propose a simple free-energy model for the morphological microstate of the cell with an active component to represent the actin cytoskeleton and a passive component for the nucleus and the remainder of the cytosol.

$$G_{cell} = F_{cyto} + F_{elas}$$

During the course of the experiment (~24 hours), the cell fluctuates between different morphological microstates while maintaining homeostasis. We impose the homeostatic constraint by ensuring that the average free-energy of the cell is maintained at a value G_s during the experiment. Minimizing the free-energy of the cell subject to this constraint yields the morphological evolution of the cell in response to its confinement.

Results

Immunofluorescence images from experiments of cells on three different line widths are shown in Fig 1b. In comparison, commonly observed morphological microstates from simulations on these line widths are depicted in Fig 1c. Note that the active and passive components of the total free-energy determine the simulated actin and vinculin 'stains' respectively.



Cell alignment is enhanced with increased confinement, as seen from the cell order parameter in Fig 1d. Interestingly, cell alignment is observed even at line widths ($w \sim 160$ μm) where stress-fiber alignment is negligible. This implies that contact guidance at wider line widths is driven purely by morphological entropy. Calculations of guidance forces exerted by the adhesive lines on the cells confirm this hypothesis.

Discussion

Cytoskeletal alignment only occurs for cells on line widths smaller than cell sizes ($w < 80$ μm) and forces arising from thermal fluctuations are negligible at all line widths. Therefore, contact guidance is predominantly driven by morphological entropy, arising from the imperfect regulation of the homeostatic process.

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References

1. Vigliotti, A., et al., (2015). *BMMB.* **15**, 761-789.